

Linkage-engineered donor–acceptor covalent organic frameworks for optimal photosynthesis of hydrogen peroxide from water and air

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Charge transfer and mass transport to catalytic sites are critical factors in photocatalysis. However, achieving both simultaneously is challenging due to inherent trade-offs and interdependencies. Here we develop a microporous covalent organic framework featuring dense donor–acceptor lattices with engineered linkages. The donor–acceptor columnar π -arrays function as charge supply chains and as abundant water oxidation and oxygen reduction centres, while the one-dimensional microporous channels lined with rationally integrated oxygen atoms function as aligned conduits for instant water and oxygen delivery to the catalytic sites. This porous catalyst promotes photosynthesis with water and air to produce H_2O_2 , combining a high production rate, efficiency and turnover frequency. This framework operates under visible light without the need of metal co-catalysts and sacrificial reagents, exhibits an apparent quantum efficiency of 17.5% at 420 nm in batch reactors and enables continuous, stable and clean H_2O_2 production in flow reactors.

Hydrogen peroxide (H_2O_2) is one of the most essential industrial products, with broad applications in biological treatment, water purification and chemical syntheses¹. It features an even higher oxidation power than high-valent metal oxidants such as KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$, functions under mild conditions and is environmentally benign to yield only water and oxygen as by-products. However, H_2O_2 is currently produced via the anthraquinone process at elevated temperature, which is energy demanding; employs palladium catalysts, high pressure H_2 and toxic solvents; and emits considerable amounts of chemical waste². Therefore, it is highly desirable to explore alternative yet efficient, mild and green approaches. In this aspect, photocatalytic H_2O_2 generation offers the advantages of energy efficiency, mild reaction conditions, high selectivity to minimize waste by-products, avoidance

of hazardous precursors and potential for continuous production. As with general photocatalysis, a problematic issue is the quick recombination of photogenerated electrons and holes, which dissipates the excitation energy as heat, resulting in a low catalytic efficiency even when facilitated by sacrificial reagents^{3–5}. To combat this problem, designing photocatalysts with a long-lived charge separation state could be an effective strategy. Ideally, this can be achieved by ensuring the timely transport and accumulation of electrons and holes, so as to facilitate their separation and to support favourable kinetics for increased redox efficiency. Similarly, the delivery of reactants and accessibility to the catalytic sites are also critical for reactions involving multiple electrons, as these conditions ensure a more uniform distribution of reactants at the catalytic sites to promote faster

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reaction kinetics. Nevertheless, accomplishing these objectives collectively is challenging^{6,7}.

While semiconductor nanomaterials such as TiO₂, ZnO and CdS have been widely studied, they generally have wide bandgaps with narrow absorption ranges, which contributes to their limited efficiencies⁸. By contrast, metal–organic frameworks present tunable structures but are moisture sensitive with risks of metal leaching⁹. Comparatively, graphitic carbon nitride (g-C₃N₄) is a popular metal-free photocatalyst but suffers from problems such as poor dispersibility, narrow absorption range and rapid electron–hole recombination¹⁰. Resorcinol–formaldehyde resins have also garnered attention for their relatively straightforward synthesis and low-cost precursors, but are often associated with low exposed catalytic site surface area and difficulty in structural modification¹¹. As a class of crystalline porous polymers, covalent organic frameworks (COFs) offer several advantages as photocatalytic materials, such as versatility, tailored energy gaps, high surface area, photochemical stability and selective catalysis^{12–18}. By integrating functional units such as (diaryl-amino)benzene¹⁹, bipyridine²⁰, heptazine²¹, benzothiadiazole²² and diazine²³, or through conjugation^{24,25} to activate and accelerate the oxygen reduction pathway, recent findings indicate that COFs exhibit promise as effective metal-free photocatalysts for H₂O₂ production. Nevertheless, a notable proportion of both inorganic and organic photocatalytic systems have demonstrated a strong dependence on pure oxygen gas as the primary feedstock, exhibiting reduced reactivity when substituted with air. Furthermore, additional alcohol sacrificial donors are frequently introduced to facilitate the oxidation half-reaction and enhance the dispersibility of the photocatalyst. If a system could utilize water, air and light for photosynthesis of H₂O₂ under ambient conditions, it could represent a noteworthy breakthrough, amalgamating sustainability, simplicity and scalability.

Herein we present key structural elements associated with the photogeneration of charge carriers, charge transport and mass delivery. In our approach, we recognize that the core of light-driven charge carrier generation involves the interplay between the electron donor (D) and acceptor (A). We demonstrate that by combining electron-rich hexaphenyl-substituted triphenylene with electron-deficient benzothiadiazole units (Fig. 1a), it is possible to achieve both charge carrier generation and catalytic site activation. In conjunction with a dense distribution of catalytic sites and π -columnar structures, COFs are shown to facilitate photoredox reactions involving multiple electrons. The one-dimensional (1D) nanochannels, adorned with hydrogen-bond-accepting atoms, also serve as docking sites, initiating immediate water and oxygen delivery to catalytic centres. The COF photocatalyst enables the coupling of water oxidation and oxygen reduction reactions, promoting the photosynthesis of H₂O₂ from water and air with a high production rate (5.7 mmol g^{−1} h^{−1}), apparent quantum efficiency (17.5% at 420 nm) and turnover frequency (4.2 h^{−1}), in both batch and flow reactors, in the absence of metals and sacrificial donors.

Results

Structural elements and two-dimensional skeletons

To facilitate photosynthesis involving multiple electrons, we designed and combined an electron-rich hexaphenyl-substituted triphenylene donor knot (Fig. 1a, D) with an electron-deficient benzothiadiazole acceptor linker (Fig. 1a, A). These species eventually serve as catalytic sites for four-electron water oxidation and two-electron oxygen reduction, respectively. This structure also creates hexavalent lattices to achieve the highest valency, as well as the densest array of catalytic sites among all topologies of COFs²⁶. To form well-suited micropores, the acceptor is elongated by incorporating two additional phenyl units, resulting in an extended terphenyl backbone. The pores are also specifically made to function as aligned conduits, where hydrogen-bond-accepting atoms such as oxygen or nitrogen lining the walls of 1D channels serve as docking sites, facilitating immediate water

and oxygen delivery through a capillary effect. To connect the D and A units into two-dimensional (2D) hexavalent lattices, non-conjugated hydrazone linkages (Fig. 1a, dashed lines), partially π -conjugated imine linkages (solid lines) and fully π -conjugated vinylenes (double lines) are engineered. These linkages bestow the hexavalent 2D D–A frameworks with the same trigonal lattice while presenting distinct molecular interfaces between D and A, featuring tuned degrees of in-plane π -conjugation (Fig. 1a).

The non-conjugated hydrazone-linked Hz-TP-BT-COF (BT, benzothiadiazole; TP, triphenylene; Fig. 1b), partially π -conjugated imine-linked Im-TP-BT-COF (Fig. 1c) and fully π -conjugated vinylenes-linked *sp*²c-TP-BT-COF (Fig. 1d) were obtained via topology-guided polymerization under solvothermal conditions (Supplementary Methods). Powder X-ray diffraction (PXRD) patterns revealed distinct peaks characteristic of hexavalently linked crystalline porous frameworks (Fig. 2a and Extended Data Fig. 1). The Hz-TP-BT-COF exhibited peaks at 2.64°, 4.56°, 5.28°, 7.00°, 7.94°, 10.64° and 25.08°, corresponding to the (100), (110), (200), (210), (300), (310) and (001) facets, respectively (Fig. 2a, blue crosses). Structural reconstruction (Fig. 2a, green curve) with sync AA stacking mode followed by Pawley refinement (Fig. 2a, red curve) yielded unit cell parameters of $a = 3.87$ nm, $b = 3.89$ nm, $c = 3.67$ Å, $\alpha = \beta = 90^\circ$ and $\gamma = 60^\circ$ (Extended Data Fig. 2a,d and Supplementary Note 1), while other stacking modes could not reproduce the experimental peak positions and intensities (Supplementary Figs. 1–3 and Supplementary Discussion). The Im-TP-BT-COF and *sp*²c-TP-BT-COF also assume similar trigonal lattices (Extended Data Fig. 2b,c,e,f and Supplementary Table 1). The atomistic coordinates and crystallographic information files (CIFs) of sync-AA-stacked COFs are available as Supplementary Data 1 and 2.

Surprisingly, Hz-TP-BT-COF with its non-conjugated hydrazone linkage can isolate the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) on D and A, respectively (Fig. 3a,d). By contrast, partially and fully π -conjugated Im-TP-BT-COF (Fig. 3b,e) and *sp*²c-TP-BT-COF (Fig. 3c,f) host a delocalized HOMO across D and A extending to the whole 2D plane. As a result, the non-conjugated Hz-TP-BT-COF (Fig. 1b) can develop each knot as the oxidation site and each linker as the reduction centre, activating the lattice into a catalytic scaffold upon irradiation to enable the use of both HOMO and LUMO energies in photosynthesis. This contrasts sharply with the donor knots, which are impeded as oxidation centres when they are integrated in the π -conjugated Im-TP-BT-COF and *sp*²c-TP-BT-COF. This difference in frontier orbitals could be decisive to the photocatalytic activity.

Stoichiometrically, the molar ratio of reduction to oxidation sites is 1:3 (Fig. 1a–e), which exceeds the requirement for electrons in the two half-reactions (1:2). This can ensure the sufficient supply of holes at the D knots for water oxidation reactions involving four electrons, a key step in photosynthesis.

This D–A architecture is unique in that each D and A unit is interfaced yet segregated to form hexavalent D and A π -columns (Fig. 1f), with a density among the highest found in COF topologies²⁶. Moreover, the ordered π -arrays are hole and electron conductive via hopping or band-like conduction owing to maximized intra-column π -cloud delocalization and minimized reorganization energy²⁷. This segregated D and A columnar alignment contributes greatly to photosynthesis as they offer pathways to transport photogenerated charges along the *z* direction and facilitate charge accumulation at the catalytic sites to empower reactions involving multiple electrons^{16–18}.

One-dimensional nanochannels and water delivery

The linkage and skeleton structures were unambiguously characterized by various analytical methods. Fourier transform infrared (FT-IR) spectroscopy revealed characteristic vibration bands of the hydrazone, imine and vinylenes linkages (Fig. 2b and Supplementary Fig. 4). Solid-state ¹³C cross-polarization magic angle spinning nuclear

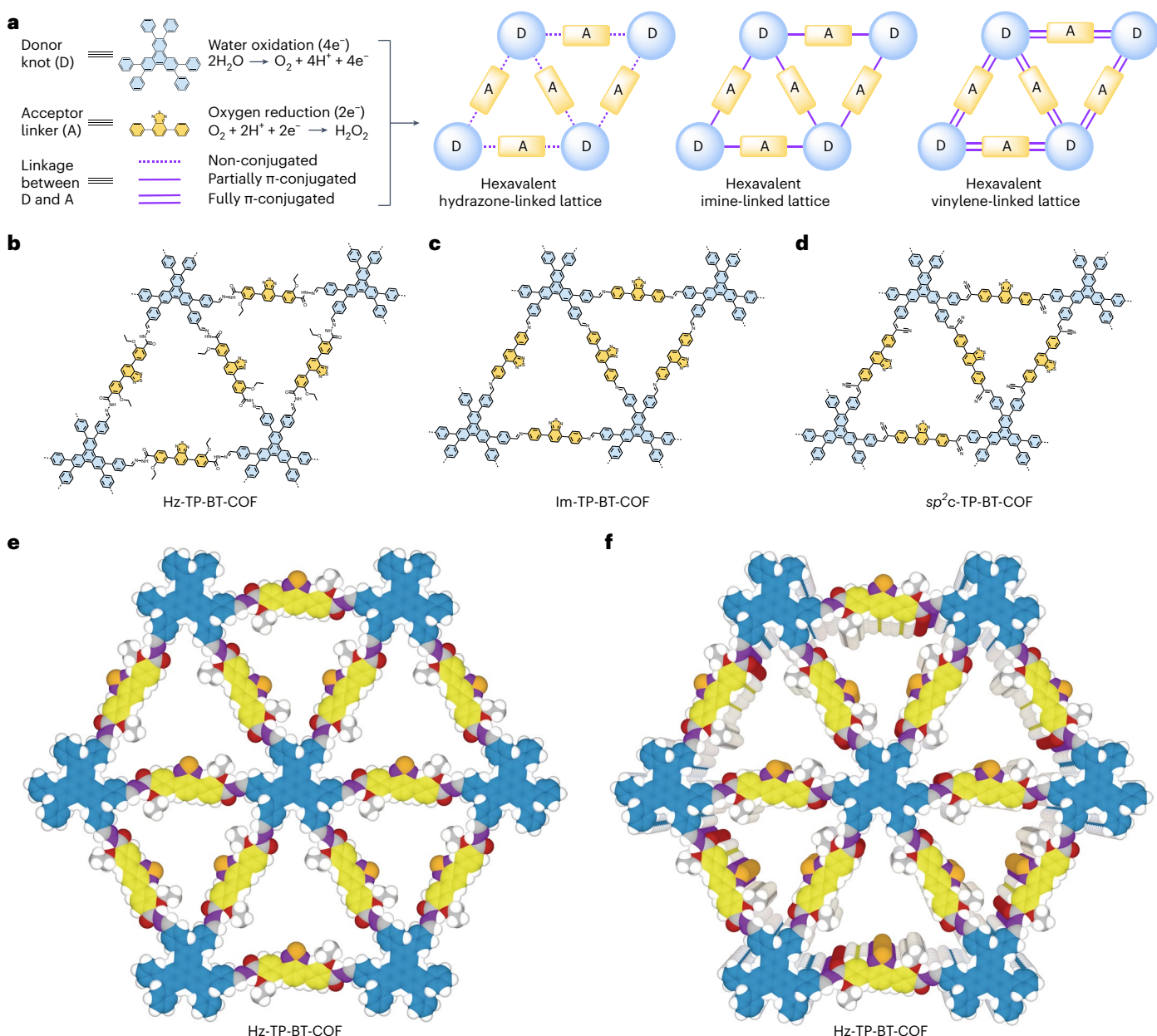


Fig. 1 | Hexavalent electron donor–acceptor COFs. **a**, Schematics of hexavalent electron donor knot (D), acceptor linker (A) and linkages between D and A; water oxidation reaction at the D site; oxygen reduction reaction at the A site; and three hexavalent D–A COFs with non-conjugated hydrazone, partially π -conjugated imine and fully π -conjugated vinylene linkages (shown with two pores). Each D and A unit serves as a catalytic site, all of which are molecularly interfaced in the hexavalent lattice to attain the highest density among COF topologies. **b–d**,

Chemical structures of non-conjugated hydrazone-linked Hz-TP-BT-COF (**b**), partially π -conjugated imine-linked Im-TP-BT-COF (**c**) and fully π -conjugated vinylene-linked sp^2c -TP-BT-COF (**d**; blue, TP donor knot; yellow, BT acceptor linker). **e, f**, Reconstructed hexavalent Hz-TP-BT-COF with one layer (**e**) and ten layers (**f**), shown with six pores (blue, yellow and grey, C; orange, S; purple, N; red, O; white, H).

magnetic resonance (CP/MAS NMR) spectroscopy confirmed the carbon atom signals from linkages, knots and linkers (Fig. 2c and Supplementary Fig. 5). Field-emission scanning electron microscopy disclosed micrometre rod-like morphologies for Hz-TP-BT-COF and rod-like objects assembled from flakes for Im-TP-BT-COF and sp^2c -TP-BT-COF (Supplementary Fig. 6). Notably, these COFs are stable over 350 °C under nitrogen (Supplementary Fig. 7).

Nitrogen sorption isotherm measurements at 77 K revealed type I isotherms (Fig. 2d and Supplementary Fig. 8), indicative of microporosity. The Hz-TP-BT-COF, Im-TP-BT-COF and sp^2c -TP-BT-COF host Brunauer–Emmett–Teller surface areas of 851 m² g^{−1}, 360 m² g^{−1} and 330 m² g^{−1}, and pore volumes of 0.33 cm³ g^{−1}, 0.17 cm³ g^{−1} and

0.15 cm³ g^{−1}, respectively (Fig. 2e). Applying non-local density functional theory to analyse pore size distributions revealed dual micropores of 1.2–1.9 nm (Fig. 2e and Supplementary Fig. 9), consistent with the sync AA stacking mode (Supplementary Fig. 1).

The well-developed pores of Hz-TP-BT-COF exhibit a sharp water uptake and release via a capillary effect at low vapour pressure and 298 K (Fig. 2f, green curve), while those of sp^2c -TP-BT-COF (blue curve) and Im-TP-BT-COF (orange curve) become sluggish and shift to higher pressure ranges. These differences are likely related to the surface exposure of three single-file chains of oxygen or nitrogen atoms in each 1D channel (Fig. 2g–i, red and blue arrows)²⁸. Supported by electrostatic potential calculations (Extended Data Fig. 3 and Supplementary

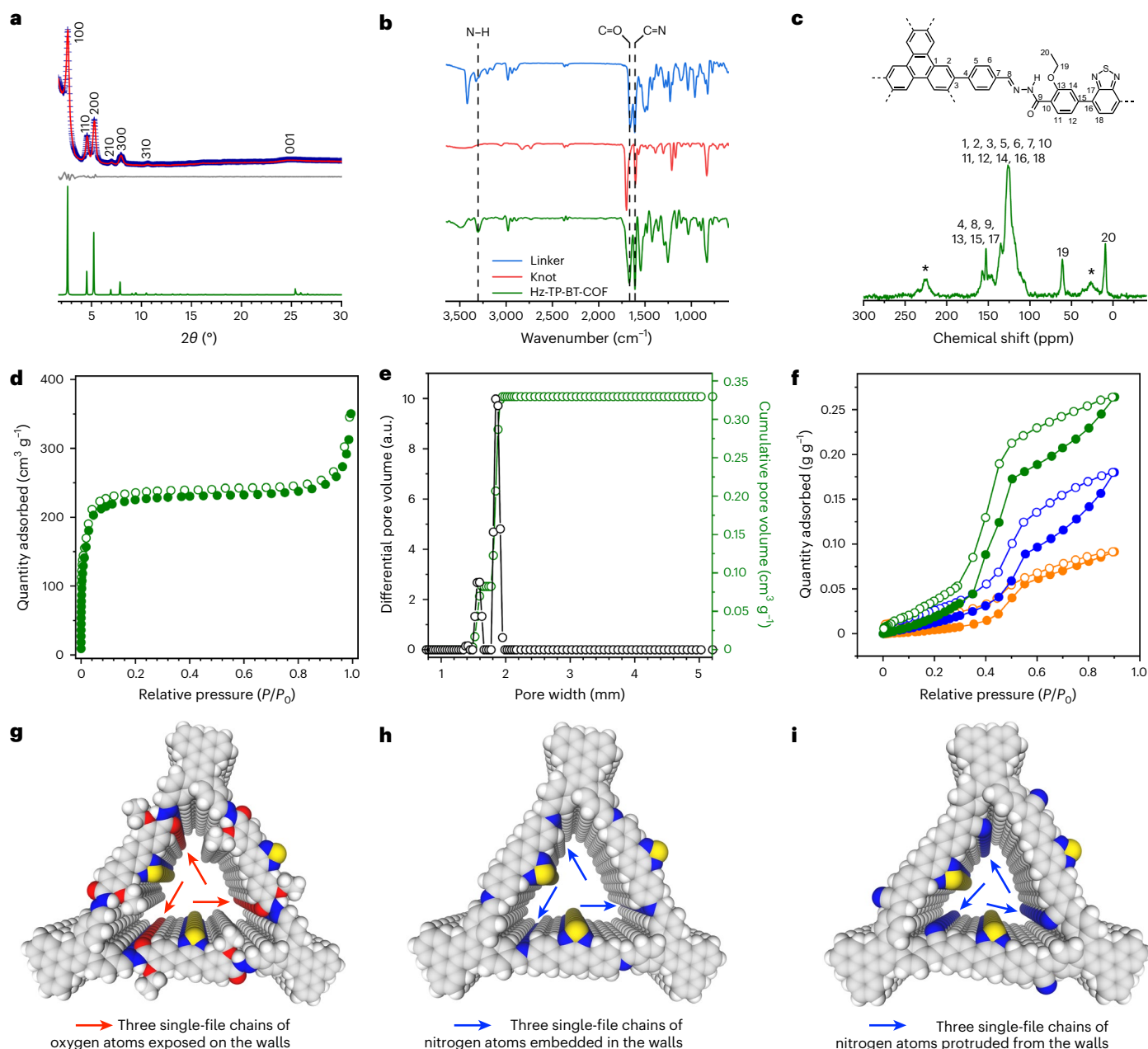


Fig. 2 | Crystal and pore structures. **a**, Experimentally obtained PXRD patterns (blue crosses), Pawley refined patterns (red curve), their differences (grey curve) and simulated sync AA stacking pattern (green curve) of Hz-TP-BT-COF. **b**, FT-IR spectra of Hz-TP-BT-COF (green curve), its knot (red curve) and its linker (blue curve). The three dashed lines from left to right show the vibration bands of N-H, C=O and C=N, respectively. **c**, Solid-state ^{13}C CP/MAS NMR spectrum of Hz-TP-BT-COF and carbon signal assignment (*, side band). **d**, Nitrogen sorption isotherms of Hz-TP-BT-COF (solid circle, adsorption; open circle, desorption). P , pressure;

P_0 , baseline pressure. **e**, Cumulative pore volume (green circles) and pore size distribution (black circles) curves for Hz-TP-BT-COF. **f**, Water vapour sorption isotherms of Hz-TP-BT-COF (green), Im-TP-BT-COF (orange) and sp^2c -TP-BT-COF (blue) at 298 K (solid circle, adsorption; open circle, desorption). **g–i**, Inner channel single-file chains of oxygen and nitrogen atoms (red and blue arrows, respectively) on pore walls, which offer hydrophilic sites for water nucleation and access to the corner nanogrooves in Hz-TP-BT-COF (**g**), Im-TP-BT-COF (**h**) and sp^2c -TP-BT-COF (**i**).

Note 2), the exposed hydrophilic chains endow the micropores with hydraulic sensitivity by offering open sites along the z direction for water nucleation and cluster formation²⁸, allowing an instant and spontaneous water supply to the corner nanogrooves of the knot water oxidation sites, and deliver dissolved oxygen to the linker oxygen reduction centres across the catalytic network.

Charge separation and energy levels

The Hz-TP-BT-COF, sp^2c -TP-BT-COF and Im-TP-BT-COF harvest visible photons, with medium optical bandgaps of 2.59 eV, 2.40 eV and 2.26 eV,

respectively (Figs. 4a and 5a, Supplementary Fig. 10 and Supplementary Note 3). Under visible light irradiation (wavelength $\lambda > 420$ nm), Hz-TP-BT-COF exhibited the lowest charge transfer resistance (Fig. 5b) and highest photocurrent density (Fig. 5c), indicating the greatest efficiency of photoinduced charge carrier generation. Among the series, the A units in the non-conjugated Hz-TP-BT-COF are more electron deficient to facilitate photoinduced electron transfer, as demonstrated by charge distribution analysis, which revealed average charges of -0.033 , -0.059 and -0.072 on the carbon atoms in the A sites of Hz-TP-BT-COF, Im-TP-BT-COF and sp^2c -TP-BT-COF, respectively (Supplementary Fig. 11

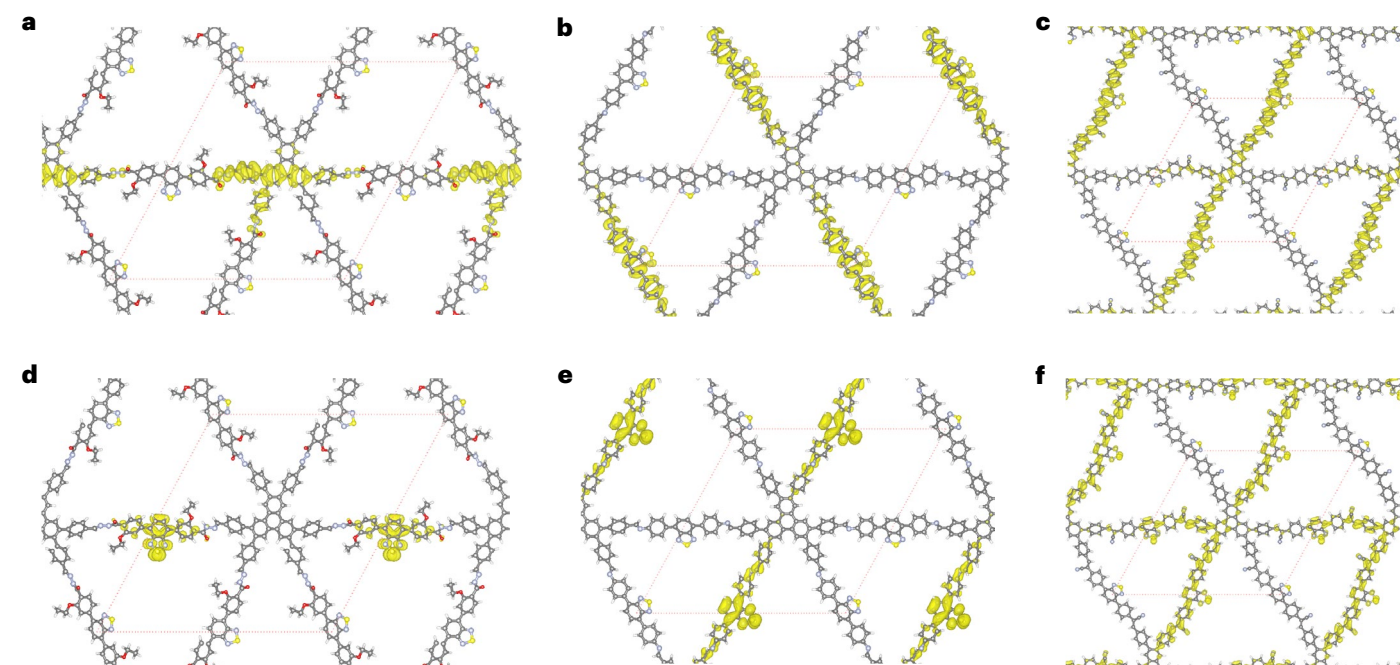


Fig. 3 | Frontier orbitals. **a–c**, The HOMO of Hz-TP-BT-COF (**a**), Im-TP-BT-COF (**b**) and sp^2c -TP-BT-COF (**c**). **d–f**, The LUMO of Hz-TP-BT-COF (**d**), Im-TP-BT-COF (**e**) and sp^2c -TP-BT-COF (**f**).

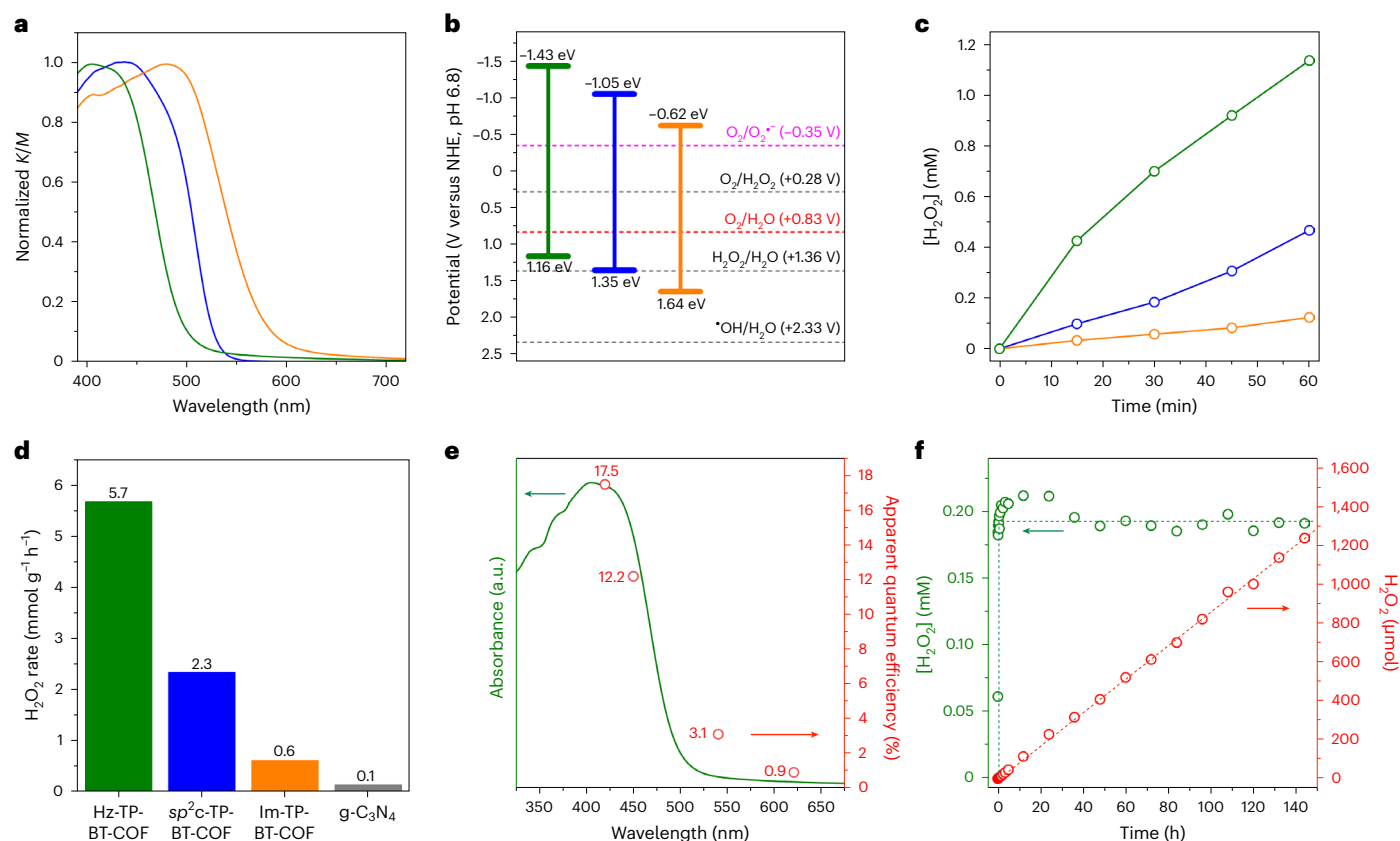


Fig. 4 | Photosynthesis with water and air. **a**, Solid-state electronic absorption spectra. K/M , Kubelka–Munk function. **b**, Experimentally derived energy band alignments of COFs compared to oxygen reduction (pink line at -0.35 V) and oxygen evolution (red line at $+0.83$ V) potentials (versus NHE at pH 6.8) along with other reactions. **c**, Batch reactor results for photosynthesis of H_2O_2 from water and air over 60 minutes under visible light ($\lambda \geq 420$ nm). In **a–c**, green, Hz-TP-BT-COF; blue, sp^2c -TP-BT-COF; orange, Im-TP-BT-COF. **d**, Batch reactor H_2O_2 production rate by COFs in comparison with that of the state-of-the-art g- C_3N_4 .

e, Apparent quantum efficiency (red circles) and electronic absorption spectrum (green curve) of Hz-TP-BT-COF. The green arrow shows the absorbance, while the red arrow shows the apparent quantum yield. **f**, Continuous production of H_2O_2 from water and air in flow reactors over 144 hours (LED, $\lambda = 460$ nm; green circles, $[H_2O_2]$ versus time; red circles, amount of H_2O_2 solution produced versus time). The green arrow shows the H_2O_2 concentration, while the red arrow shows the H_2O_2 amount.

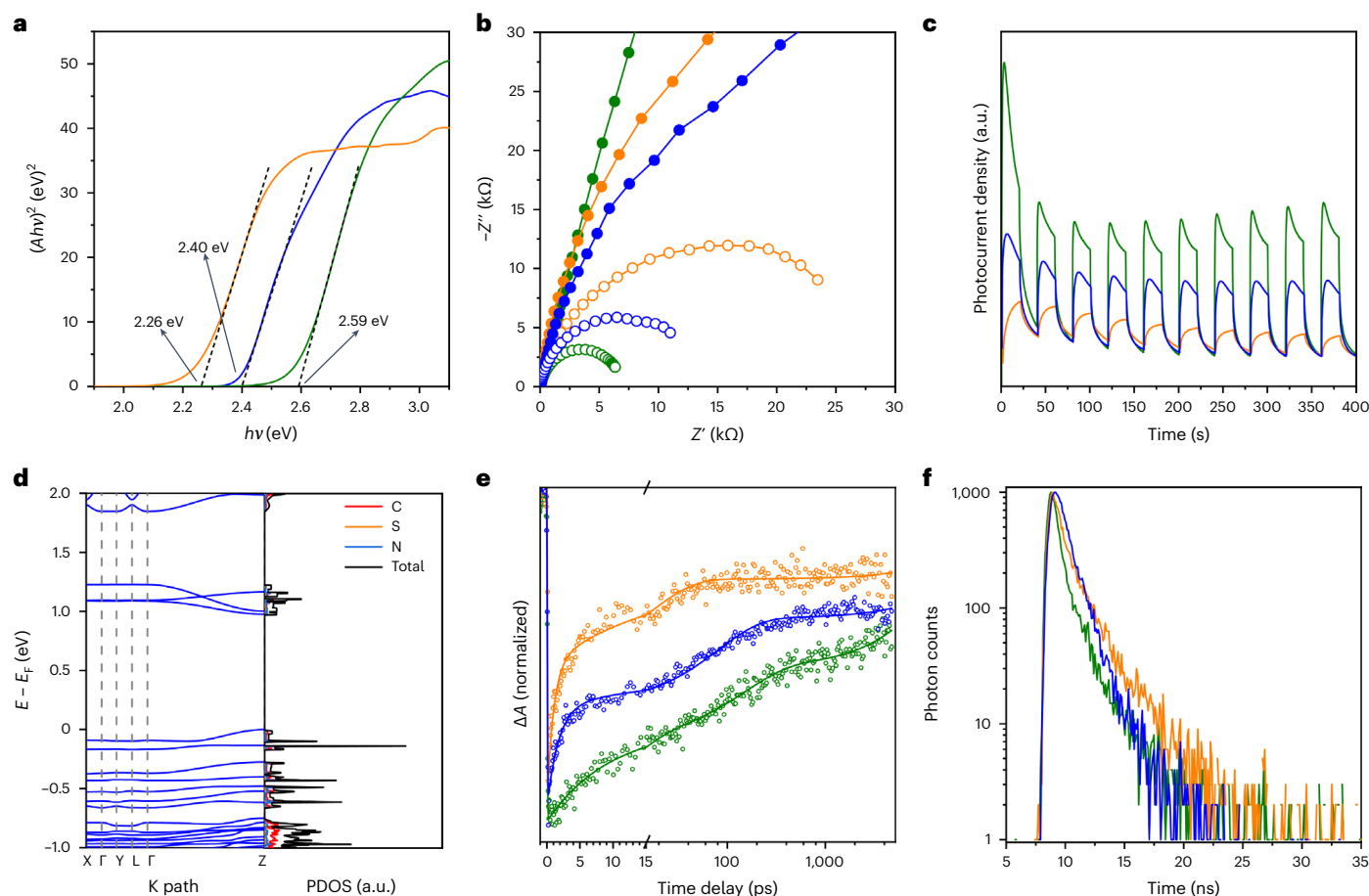


Fig. 5 | Photophysical and electrochemical measurements. **a**, Tauc plot (values with arrows are the optical bandgap). A , absorption coefficient; h , Planck constant; ν , frequency. **b**, Electrochemical impedance spectra (solid circle, in the dark; open circle, under visible light irradiation). Z' , real axis; Z'' , imaginary axis.

c, Photocurrent density measurements. **d**, Band structure and projected density of states (PDOS) for Hz-TP-BT-COF. E , energy; E_F , Fermi energy. **e**, Transient spectroscopy kinetics trace plots. **f**, Fluorescence decay profile. In **a–c**, **e** and **f**, green, Hz-TP-BT-COF; orange, Im-TP-BT-COF; blue, sp^2 c-TP-BT-COF.

and Supplementary Note 4). Moreover, the D knots in Hz-TP-BT-COF are more electron rich, as evidenced by their much denser p -orbital population of electrons near the valence band edge (Fig. 5d and Supplementary Fig. 12). These two beneficial effects render Hz-TP-BT-COF able to trigger ultrafast photoinduced electron transfer from D to A and charge separation. We investigated the photoexcited state dynamics by conducting transient absorption (TA) measurements. Within 0.4 ps, the TA spectra show the formation of a negative feature, which can be assigned to the combined ground state bleaching and stimulated emission. Simultaneously, a broad positive feature that extends to the infrared region can be attributed to photoinduced absorption (Supplementary Fig. 13). TA measurement revealed an average lifetime of charge separated state (τ_{cs}) of 75 ± 2 ps for Hz-TP-BT-COF, which is much longer than those of Im-TP-BT-COF (7.4 ± 0.4 ps) and sp^2 c-TP-BT-COF (38 ± 2 ps; Fig. 5e and Supplementary Table 2). This suggests a better charge separation capability with reduced recombination rates for Hz-TP-BT-COF. Correspondingly, Hz-TP-BT-COF displayed a shorter fluorescence lifetime of 1.18 ns compared to Im-TP-BT-COF (1.70 ns) and sp^2 c-TP-BT-COF (1.27 ns; Fig. 5f). As the non-conjugated 2D network is likely inactive for in-plane conduction, the prominent photocurrent obtained in Hz-TP-BT-COF most likely originates from the vertical charge transport and accumulation of photogenerated charges across the π -columns^{16–18}. This is further supported by the low charge transfer resistance and long lifetime of the charge separation state.

The Hz-TP-BT-COF (Fig. 4b, green plot, Supplementary Fig. 14 and Supplementary Note 5) possesses the most negative conduction band

level at -1.43 V (versus normal hydrogen electrode (NHE), pH 6.8), followed by sp^2 c-TP-BT-COF (blue plot, -1.05 V) and Im-TP-BT-COF (orange plot, -0.62 V), which suggests the greatest photoreduction ability of the A sites in Hz-TP-BT-COF to drive the oxygen reduction reaction into H_2O_2 via the superoxide anion intermediate (O_2^- , dashed pink line, -0.35 V). On the other hand, the valence band levels of Hz-TP-BT-COF, sp^2 c-TP-BT-COF and Im-TP-BT-COF are $+1.16$, $+1.35$ and $+1.64$ V (versus NHE, pH 6.8), respectively, which are positive enough to allow the D sites to catalyse water oxidation involving four electrons (dashed red line, $+0.83$ V; Supplementary Fig. 15 and Supplementary Discussion).

With the hexavalent 2D lattice, we created D–A frameworks that exhibit unique structures and properties that contrast sharply with those obtained with traditional strategies^{19–25}. We anticipate that merging these distinct structural elements into one photocatalyst could enable efficient photosynthesis with water and air.

Batch reactors for photosynthesis

Batch reactors were fabricated by dispersing Hz-TP-BT-COF in water to prepare stable homogeneous dispersions (Supplementary Figs. 16 and 17 and Supplementary Note 6). Notably, upon irradiation with visible light ($\lambda = 420–600$ nm) under air bubbling at 20°C , Hz-TP-BT-COF spontaneously initiates photosynthesis with water and air and generates H_2O_2 , without showing any induction period. The H_2O_2 concentration exceeds 1.14 mM upon 60 min irradiation (Fig. 4c, green curve). By contrast, sp^2 c-TP-BT-COF (blue curve) and Im-TP-BT-COF (orange curve) yielded a concentration of only 0.47 and 0.12 mM, respectively.

Consequently, Hz-TP-BT-COF achieves a H_2O_2 production rate as high as $5.7 \text{ mmol g}^{-1} \text{ h}^{-1}$ (Fig. 4d, green bar), which is much higher than those of $\text{sp}^2\text{c-TP-BT-COF}$ (blue bar, $2.3 \text{ mmol g}^{-1} \text{ h}^{-1}$) and Im-TP-BT-COF (orange bar, $0.6 \text{ mmol g}^{-1} \text{ h}^{-1}$). During the reaction, the pH of the aqueous solution was also found to decrease slightly due to the dissociation of H_2O_2 (Supplementary Fig. 18 and Supplementary Discussion). Despite possible competing side reactions such as nitrogen and carbon dioxide photoreduction from the introduction of air and the hydrogen evolution from water (Supplementary Fig. 19 and Supplementary Discussion), we performed an extensive analysis of the gaseous and aqueous phases and found no evidence of any by-products (Supplementary Figs. 20 and 21). Furthermore, contamination by Pd was ruled out (Supplementary Table 3 and Supplementary Discussion), which further highlighted the superior selectivity of Hz-TP-BT-COF for H_2O_2 production.

Replacing air-saturated water with oxygen-saturated water increases the production rate to $6.5 \text{ mmol g}^{-1} \text{ h}^{-1}$ (Supplementary Fig. 22), while adding sacrificial donors such as benzyl alcohol enhances the rate to $7.1 \text{ mmol g}^{-1} \text{ h}^{-1}$ (Supplementary Fig. 23). On the contrary, the amorphous analogue Hz-TP-BT without the ordered D–A structure (Supplementary Methods) greatly decreases the H_2O_2 production rate to only $0.9 \text{ mmol g}^{-1} \text{ h}^{-1}$. Notably, Hz-TP-BT-COF drastically outperforms the state-of-the-art $\text{g-C}_3\text{N}_4$ photocatalyst²⁹ by more than 50-fold (Fig. 4d, grey bar) and TPB-DMTP-COF³⁰ by 5.6-fold (Supplementary Methods) under identical conditions. The Hz-TP-BT-COF achieves a high apparent quantum efficiency of 17.5% at 420 nm (Fig. 4e, red circle). Remarkably, its performance in terms of H_2O_2 production rate, concentration, apparent quantum efficiency (Supplementary Note 7) and turnover frequency exceeded benchmark organic heterogeneous photocatalysts, including $\text{g-C}_3\text{N}_4$ (refs. 29,31), linear polymers³², COFs^{19–25}, covalent triazine frameworks³³, covalent heptazine frameworks³⁴, polymer resins^{11,35} and self-assembled frameworks³⁶ (Supplementary Table 4). It also outperforms most inorganic materials, such as metal–organic frameworks^{9,37}, TiO_2 -based materials⁸, metal organic cages³⁸, chalcogenides³⁹ and heterojunctions constructed from various organic and inorganic materials^{40,41} (Supplementary Table 4). Considering that most reported photocatalytic systems deploy pure oxygen as feedstock, this Hz-TP-BT-COF using air directly for efficient and high-rate H_2O_2 production is truly game-changing.

Under air upon irradiation with the full wavelength ($\lambda > 300 \text{ nm}$) of the xenon lamp, the batch reactors further boost the H_2O_2 production to a rate of $12.5 \text{ mmol g}^{-1} \text{ h}^{-1}$ (Supplementary Fig. 24). Surprisingly, under ambient outdoor conditions (Supplementary Fig. 25 and Supplementary Note 8), the exposure of a water dispersion of Hz-TP-BT-COF to natural sunlight produces a prominent amount of H_2O_2 at a high rate of $2.7 \text{ mmol g}^{-1} \text{ h}^{-1}$ (Supplementary Fig. 24). The Hz-TP-BT-COF is robust under various conditions, such as in organic solvents, water and even a strong acid or base (Supplementary Fig. 26). The Hz-TP-BT-COF is stable for repetitive use with retained catalytic activity (Supplementary Fig. 27 and Supplementary Note 9) and structural integrity (Supplementary Fig. 26).

Continuous photosynthesis in flow reactors

Continuous photosynthesis with air-saturated water was achieved with flow-type photocatalytic converters. They were fabricated by connecting a flow pump with a column of Hz-TP-BT-COF (90 mg, 72 mm in length) and a collection reservoir (Supplementary Fig. 28 and Supplementary Note 10). Interestingly, this system, upon the flow of air-saturated water at a rate of 0.75 ml min^{-1} , produces H_2O_2 continuously and efficiently on irradiation with blue light (light-emitting diode (LED), $\lambda = 460 \text{ nm}$), to achieve a concentration of 0.2 mM immediately (Fig. 4f, green curve). This H_2O_2 concentration translates to the continuous transformation of about 80% dissolved oxygen⁴² into H_2O_2 . Remarkably, 6,480 millilitres of H_2O_2 (Supplementary Fig. 29) with a steady concentration were produced within 144 hours (Fig. 4f, red curve). This long-term stability highlights a distinct feature of Hz-TP-BT-COF

flow reactors, contrasting sharply with the deteriorated or saturated performance more commonly obtained in photocatalytic systems so far reported (Supplementary Table 4). Analysis with gas chromatography coupled with mass spectroscopy (Supplementary Fig. 30 and Supplementary Note 11) and ^1H NMR (Supplementary Fig. 21) confirms that the H_2O_2 solution is free of any by-products. FT-IR (Supplementary Fig. 31), PXRD (Supplementary Fig. 32), nitrogen sorption isotherms (Supplementary Figs. 33 and 34) and gravimetric measurements (Supplementary Fig. 35 and Supplementary Note 12) revealed negligible structural and weight changes after the continuous flow reaction. These results show that Hz-TP-BT-COF might provide a green and sustainable way to manufacture H_2O_2 from water and air under visible light and ambient conditions.

Photosynthetic processes

Replacement of air with argon suppresses H_2O_2 production, and negligible H_2O_2 is produced in the dark (Supplementary Fig. 36). Spin trap experiments using a nitro blue tetrazolium test for electronic absorption spectroscopy (Supplementary Fig. 37 and Supplementary Note 13) and 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO) for electron paramagnetic resonance spectroscopy revealed the formation of a superoxide radical anion ($\text{O}_2^{\cdot-}$) during photosynthesis (Supplementary Fig. 38). The absence of singlet oxygen ($^1\text{O}_2$) phosphorescence at 1,270 nm (Supplementary Fig. 39) and the negligible effect of the 1,4-diazabicyclo[2.2.2]octane $^1\text{O}_2$ scavenger excludes the likelihood of $^1\text{O}_2$ involvement (Supplementary Fig. 36). These results confirm the oxygen reduction into H_2O_2 involving $\text{O}_2^{\cdot-}$ (Fig. 4b, dashed pink line at -0.35 V). The O_2 adsorption calculations (Fig. 6a–c) also revealed more favourable adsorption energies on the BT (0.50 eV; Fig. 6c) and neighbouring phenyl (Ph; 0.56 eV; Fig. 6b) units of Hz-TP-BT-COF⁴³. Furthermore, Gibbs free energy calculations (Fig. 6d, Extended Data Fig. 4a,b, Supplementary Note 14 and Supplementary Discussion) disclosed the lowest overpotential for the $^*\text{OOH}$ intermediate on BT units of Hz-TP-BT-COF as compared to Im-TP-BT-COF and $\text{sp}^2\text{c-TP-BT-COF}$, suggesting a smaller energy barrier for the oxygen reduction reaction.

By contrast, the addition of *tert*-butanol as the hydroxyl radical ($\text{OH}^{\cdot}$) scavenger hardly affects the H_2O_2 production rate (Supplementary Fig. 36), excluding the possibility of an $\text{OH}^{\cdot}$ -mediated water oxidation process (Fig. 4b, dashed grey line at $+2.33 \text{ V}$), a result consistent with the prediction from the energy diagram. No H_2O_2 was detected in the presence of NaIO_3 as an electron scavenger (Supplementary Fig. 36), precluding the one-step direct water oxidation process (Fig. 4b, dashed grey line at $+1.36 \text{ V}$). Consequently, the D knot promotes water oxidation via an oxygen evolution process, which has a more negative redox potential (Fig. 4b, dashed red line at $+0.83 \text{ V}$). An isotopic labelling experiment using H_2^{18}O (Supplementary Figs. 40 and 41 and Supplementary Note 15) followed by headspace analysis unveiled the formation of $^{18}\text{O}_2$, further confirming that the oxygen evolution reaction is involved in the photocatalytic cycle. From Gibbs free energy calculations (Fig. 6e, Extended Data Fig. 4c,d and Supplementary Discussion), the oxygen evolution reaction is expected to take on the lower-energy-demanding dual-site ($^*\text{O}^*\text{OH}$) process, rather than the single-site ($^*\text{OOH}$) pathway on the neighbouring phenyl units of the TP core.

To further support the formation of reactive intermediates as well as the photocatalytic pathway, we performed *in situ* diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) on Hz-TP-BT-COF (Fig. 6f–h) under an oxygen and steam environment. Noticeably, the peak at $2,810 \text{ cm}^{-1}$ corresponding to O–H bending of H_2O_2 (Fig. 6f) gradually increased with photoirradiation time, suggesting the formation of H_2O_2 (refs. 23,44,45). Peaks originating from C=C stretching modes of the benzene ring ($1,603$ and $1,491 \text{ cm}^{-1}$) and BT unit ($1,550 \text{ cm}^{-1}$) also varied in intensity with photoirradiation time (Fig. 6g), suggesting their involvement as active sites in the reaction pathway^{46–48}. Peaks corresponding to *endo*-peroxide ($1,209 \text{ cm}^{-1}$), $\text{O}_2^{\cdot-}$ ($1,145 \text{ cm}^{-1}$) and O–O (886 cm^{-1}) also increased in intensity (Fig. 6h),

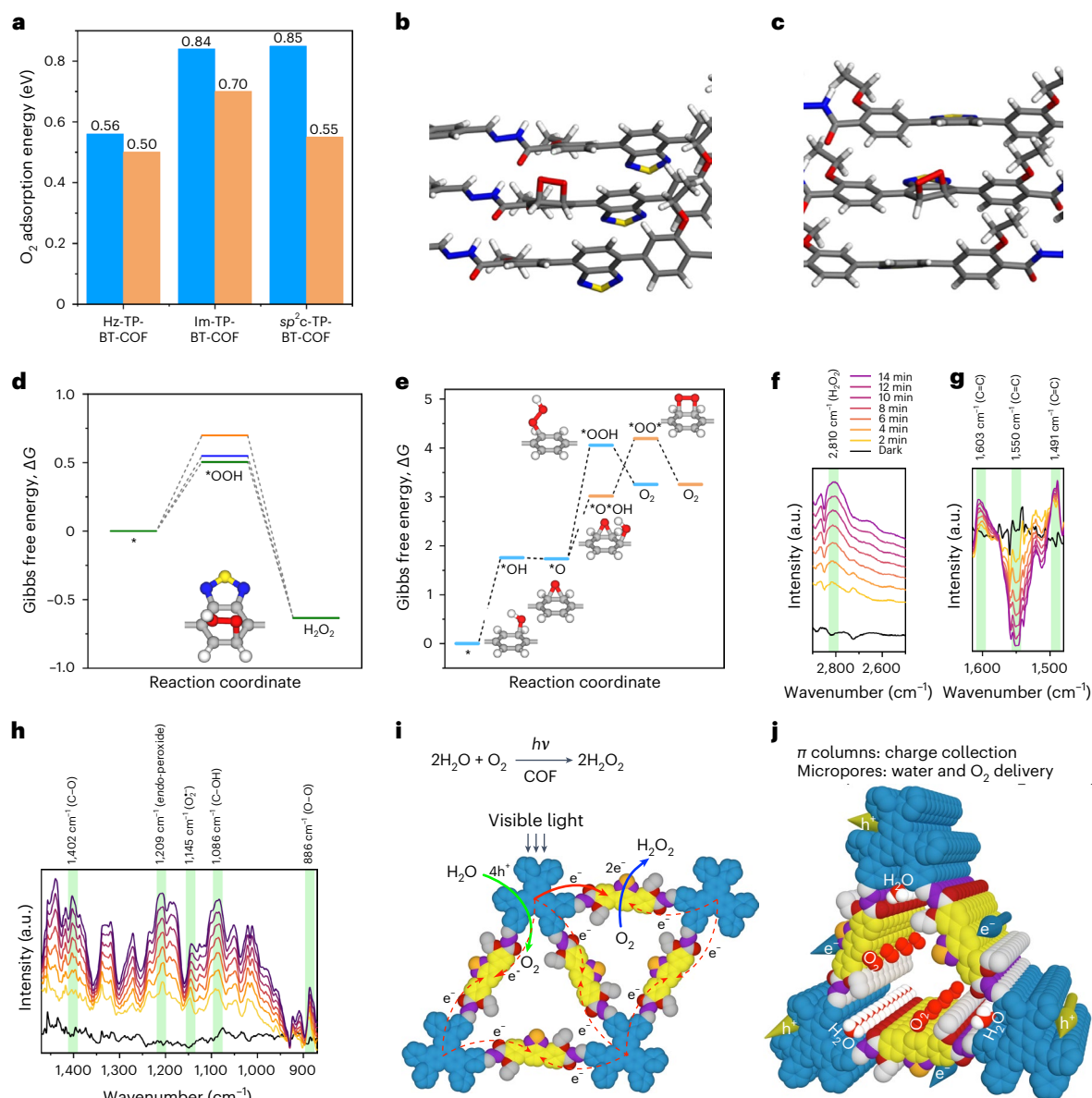


Fig. 6 | Reaction processes and pathways. **a**, O₂ adsorption energies on Ph (blue bar) and BT (orange bar) units of the COFs. **b, c**, O₂ adsorption sites on Ph (**b**) and BT (**c**) units of 1 × 1 × 3 supercell model of Hz-TP-BT-COF. **d**, Gibbs free energy diagram for oxygen reduction reaction over BT units of Hz-TP-BT-COF (green line), Im-TP-BT-COF (orange line) and sp²c-TP-BT-COF (blue line). **e**, Gibbs free energy diagram for oxygen evolution reaction via single-site (blue line) and dual-site (orange line) of Hz-TP-BT-COF. **f–h**, In situ DRIFTS study of Hz-TP-BT-COF collected at different photoirradiation times under O₂ and steam environment, from 2,900–2,500 cm⁻¹ (**f**), 1,615–1,480 cm⁻¹ (**g**) and 1,470–870 cm⁻¹ (**h**). **i**, Overall reaction scheme and one-layer hexavalent Hz-TP-BT-COF to depict the photoinduced electron transfer (red arrow) from the D knot to A linker;

water oxidation reaction involving four holes (h⁺) at the D knot (green arrow); and oxygen reduction reaction involving two electrons to produce H₂O₂ at the A site (blue arrow). Dashed arrows show six pathways for each D knot to trigger photoinduced electron transfer to the neighbouring A linkers to complete light harvesting, or photoinduced electron transfer and charge separation to produce a photogenerated hole and electron within 0.4 ps. **j**, Ordered molecular pathways for charge and mass transport (shown in one trigon of ten layers): the A π -columns (yellow) for the delivery of photogenerated electrons (e⁻) and the D π -columns (blue) for the collection of holes (h⁺), and the micropore channel to deliver water molecules (H₂O) to the corner nanogrooves of the D knots and oxygen (O₂) to the A linkers.

indicating the adsorption of O₂ and occurrence of two-step oxygen reduction to produce the superoxide radical anion^{20,23,49,50}. Finally, signals belonging to C–O (1,402 cm⁻¹) and C–OH (1,086 cm⁻¹) confirm the formation of an *OH species (Fig. 6h), the first step in the oxygen evolution reaction^{20,51}. Based on the evidence above and combining results from theoretical predictions (HOMO and LUMO distributions), we can confirm that the oxygen reduction and oxygen evolution reactions occur at the BT ring and neighbouring phenyl units of the TP core. Additional control experiments further confirmed the involvement of both O₂ and H₂O, as well as quantified their stoichiometric ratio

to be close to the theoretical value of 1:2 (Supplementary Fig. 42 and Supplementary Discussion).

This Hz-TP-BT-COF promotes photosynthesis with water and air; it enables not only ultrafast photoinduced charge separation (Fig. 6i, red arrow, and Supplementary Fig. 13a) and collection (Fig. 6j) but also an instant water and oxygen supply to the catalytic sites (Fig. 6j). The photogenerated electrons (e⁻) pool in the A columns (Fig. 6j), where H₂O₂ production (Fig. 6i, blue arrow; O₂ + 2H⁺ + 2e⁻ → H₂O₂) is promoted, while the holes accumulate in the D columns (Fig. 6j) to trigger water oxidation (Fig. 6i, green arrow; 2H₂O → O₂ + 4H⁺ + 4e⁻),

thereby driving the completion of the photosynthetic cycle (Fig. 6i; $2\text{H}_2\text{O} + \text{O}_2 \rightarrow 2\text{H}_2\text{O}_2$).

Conclusions

This work creates a unique D–A hexavalent framework for catalysing the photosynthesis of H_2O_2 from water and air, which is obtained by a topology-directed one-pot polycondensation reaction. Our investigations provide essential structural elements for the design of photocatalysts: molecularly interfaced D and A for ultrafast photoinduced electron transfer; spatially separated D and A to allow charge separation; energetically matched frontier-orbital-localized D and A to enable each D and A unit to function as oxidation and reduction active sites; and segregated D and A columnar π -arrays to offer pathways for transport and the harvest of photogenerated charges, in conjunction with well-defined hydrophilic 1D micropores for reactant delivery. This unique porous framework with predesigned pathways for both charge transport and mass supply enables photosynthesis with water and air at ambient temperature and pressure, without the need of any metal co-catalysts or sacrificial reagents. The results obtained here could be used in a broader strategy for catalytic materials, by reticulating an atomically programmed π -architecture for charge supply and tailored nanospace for pinpoint substrate delivery and product release.

Methods

Chemicals

All starting materials and solvents, unless otherwise specified, were purchased from Merck, Tokyo Chemical Industry and BLDpharm and used without prior purification.

Powder X-ray diffraction

PXRD measurements were performed on a Bruker D8 Focus Powder X-ray Diffractometer using $\text{Cu K}\alpha$ radiation (40 kV, 40 mA) at room temperature by depositing powder on a PMMA specimen holder, from $2\theta = 1.5^\circ$ to 30° (2θ , the angle between the transmitted beam and reflected beam) with a 0.01° increment.

Nitrogen sorption

Nitrogen sorption isotherms were measured at 77 K with a Micromeritics Instrument Corporation model 3Flex surface characterization analyser. The Brunauer–Emmett–Teller method was used to calculate the specific surface areas. By using the non-local density functional theory model, the pore volume was derived from the sorption curve.

Vapour sorption

The vapour sorption experiment was conducted on a Quantachrome Instruments model iQ3 equipped with manifold, vapour source and circulating water bath. Samples were pretreated under vacuum at 120°C for 2 hours. The water adsorption isotherms were performed at 298.15 K, where the saturated vapour pressure is 23.8 torr. The unit of measured volumetric vapour adsorption quantity was converted from cubic centimetres per gram to grams per gram with an equation of $w = (V/22,414) \times 18.020$, where w is weight and V is volume.

Fourier transform infrared spectroscopy

FT-IR spectra were recorded on a Bruker VERTEX 80v spectrometer using KBr pellets.

Thermogravimetric analysis

Thermogravimetric analysis data were collected using a TA Instruments Q500 by heating from 40 to 800°C with the rate of $10^\circ\text{C min}^{-1}$ under nitrogen flow (10 ml min^{-1}).

Solution-state NMR spectroscopy

The ^1H and ^{13}C NMR spectra were acquired on a Bruker AVANCE I 500 MHz NMR spectrometer.

Solid-state NMR spectroscopy

Solid-state ^{13}C CP/MAS NMR spectra were measured on an AVNEO400 solid NMR spectrometer.

Matrix-assisted laser desorption–ionization time of flight spectrometry

Matrix-assisted laser desorption–ionization time of flight spectra were acquired on a Bruker Autoflex III TOF/TOF mass spectrometer.

Electrochemical measurements

The electrochemical impedance spectra, Mott–Schottky plots and photocurrent measurements were recorded on a CHI760D electrochemical workstation with a standard three-electrode system comprising a COF-coated fluorine-doped tin oxide glass working electrode, Pt wire counter electrode and Ag/AgCl reference electrode. A Na_2SO_4 solution (0.2 M, pH = 6.8) was used as the electrolyte and purged with argon before measurements. A 300 W xenon lamp with a 420 nm cut-off filter was used for photocurrent measurements and impedance measurements under irradiation.

Electron paramagnetic resonance spectroscopy

Electron paramagnetic resonance data were obtained by a JEOL JES-FA200 electron paramagnetic resonance spectrometer using DMPO (0.1 M) as the spin trap, which forms an adduct with the ' O_2^- ' produced during photoirradiation (30 W blue LED, $\lambda = 460\text{ nm}$) in the presence of photocatalysts. The data were collected on the measurement parameters (scanning frequency, 9,050 MHz; scanning power, 5 mW; central field, 323 mT; scanning width, 100 G (Gauss); scanning temperature, 298 K).

Electronic absorption spectroscopy

Electronic absorption spectroscopy was conducted on a JASCO model V-570 spectrophotometer.

Field-emission scanning electron microscopy

Field-emission scanning electron microscopy images were taken on a JEOL model JSM-6701F operating at an accelerating voltage of 5.0 kV. The samples were prepared by drop-casting isopropanol suspension onto a silica wafer and were coated with gold.

Transient absorption spectroscopy

TA spectroscopy was conducted with a HELIOS femtosecond TA spectrometer (Ultrafast Systems). The 400 nm (3.1 eV) pump pulses were generated by frequency doubling of the 800 nm fundamental output from the regenerative amplifier (Libra, 1 kHz, 50 fs) with a beta barium borate crystal. The probe beam was generated by focusing a part of the 800 nm fundamental output from the regenerative amplifier onto a 2-mm-thick sapphire plate or 12.7-mm-thick CaF_2 plate for visible and near-infrared spectral windows, respectively. The probe white light beam was passed through a 750 nm short-pass filter to eliminate any residual 800 nm fundamental components to prevent any strong secondary photoexcitation of the sample. The solution samples were dissolved in a quartz cuvette (2 mm path length) and constantly stirred using a magnetic stirrer during the experiments to prevent thermal lensing and sample degradation. Kinetics traces at selected wavelengths are 470 nm for Hz-TP-BT-COF and sp^2c -TP-BT-COF and 670 nm for Im-TP-BT-COF.

Time-resolved fluorescence spectroscopy

The decay curves were conducted on a Hamamatsu model compact fluorescence lifetime spectrometer C11367 (QuantaTaurus-Tau) that contains an LED pulsed light source, monochromator, photomultiplier tube, iris (aperture) for adjusting light level and filter for cutting excitation light. The COF samples were dispersed in water and irradiated with an internal LED at $\lambda = 405\text{ nm}$.

Photoluminescence spectroscopy

Photoluminescence spectra were recorded with an Edinburgh FLS1000 spectrofluorometer equipped with a Hamamatsu near-infrared photomultiplier detector (C9940-02), in conjunction with either a continuous-wave 808 nm diode laser (MDL-III-808 nm) or 980 nm diode laser (MDL-III-980 nm).

Elemental analysis

Elemental analysis was recorded on a Thermo Fisher Scientific FlashSmart CHNS Elemental Analyser by weighing the samples in tin containers and then combusting at high temperature with oxygen.

In situ DRIFTS

DRIFTS measurements were performed on a Nicolet iS50 FT-IR instrument. The photocatalyst was filled into an in situ infrared holder in the chamber. Before the measurement, the chamber was degassed under argon flow at a rate of 20 ml min⁻¹ for 1 h. The baseline was then obtained at room temperature. Then, oxygen and water vapour were purged into the chamber for 1 h under dark conditions. Irradiation under visible light through the window of the chamber was initiated, and the spectra were recorded at specific intervals.

Inductively coupled plasma mass spectrometry

The metal (Pd) content was determined by inductively coupled plasma mass spectrometry on an Agilent 7700x analyser. The sample was first digested with HNO₃/HCl/HF (3:1:1) in a microwave at 240 °C for 15 minutes and topped up to 10 ml with H₂O. Clear solution was observed prior to analysis.

Computational details

Structural simulations were conducted using the self-consistent-charge density-functional tight-binding method (SCC-DFTB; ref. 52) as implemented in the Amsterdam Modelling Suite (AMS), version DFTB 2021 (ref. 53). The 3ob-3-1 parameter set for the X–Y element pair interactions (X, Y = C, H, N, O and S) was used⁵⁴, coupled with an empirical dispersion correction of the universal force field⁵⁵. The periodic models of the materials were obtained using full structural optimization, including the unit cell shape and volume. Higher precision optimizations for selected structures as well as all electronic property calculations were then performed using density functional theory calculations using the projector augmented-wave⁵⁶ method and Perdew–Burke–Ernzerhof⁵⁷ exchange–correlation functional, with Grimme's D3 approach to the London dispersion correction⁵⁸. During the structure optimization, we employed a cut-off energy of 400 eV for the plane-wave basis set and set 0.01 eV Å⁻¹ as the force convergence criterion. For static calculations, the cut-off energy was set to 600 eV. In the self-consistent field iteration, we used the energy convergence criterion of 10⁻⁵ eV and 10⁻⁶ eV for the optimization and the static calculation, respectively. We used a κ -mesh (which is a set of points in the Brillouin zone used to sample the electronic structure of a material) of 1 × 1 × 4 and 1 × 1 × 8 for the structural relaxation and the converged charge density. To simulate the H₂O₂ generation reaction and the oxygen evolution reaction, we constructed a 1 × 1 × 3 supercell of the COFs. And we applied a computational hydrogen electrode model to derive the Gibbs free energies with Nørskov's approach⁵⁹.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding author on request. The atomistic coordinates for the final optimized structures are provided as Supplementary Data 1.

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Author contributions

D.J. conceived the idea and led the project. R.L. and Y.C. conducted the experiments and measurements. H.Y., M.P. and T.H. performed the computational calculations. Y.G. and T.C.S. conducted and analysed the TA measurements. R.L., Y.C., H.Y., M.P., Y.G., T.C.S., T.H. and D.J. interpreted the results, and R.L., Y.C. and D.J. wrote the paper. All authors read and commented on the paper.

Competing interests

The authors declare no competing interests.

Additional information

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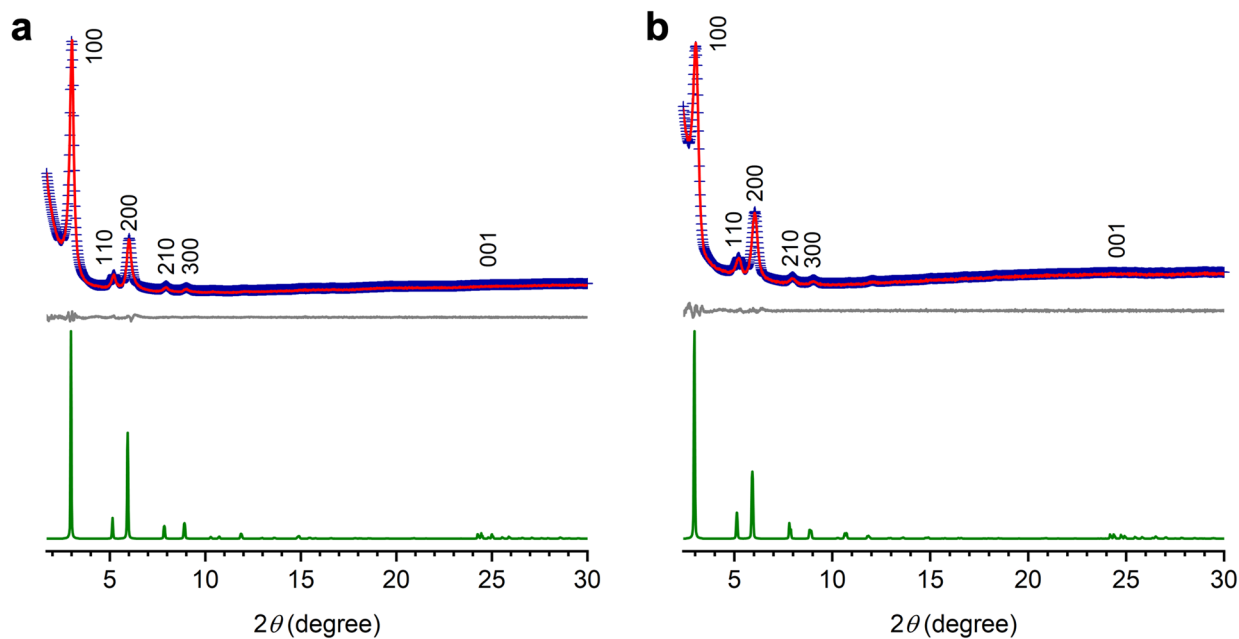
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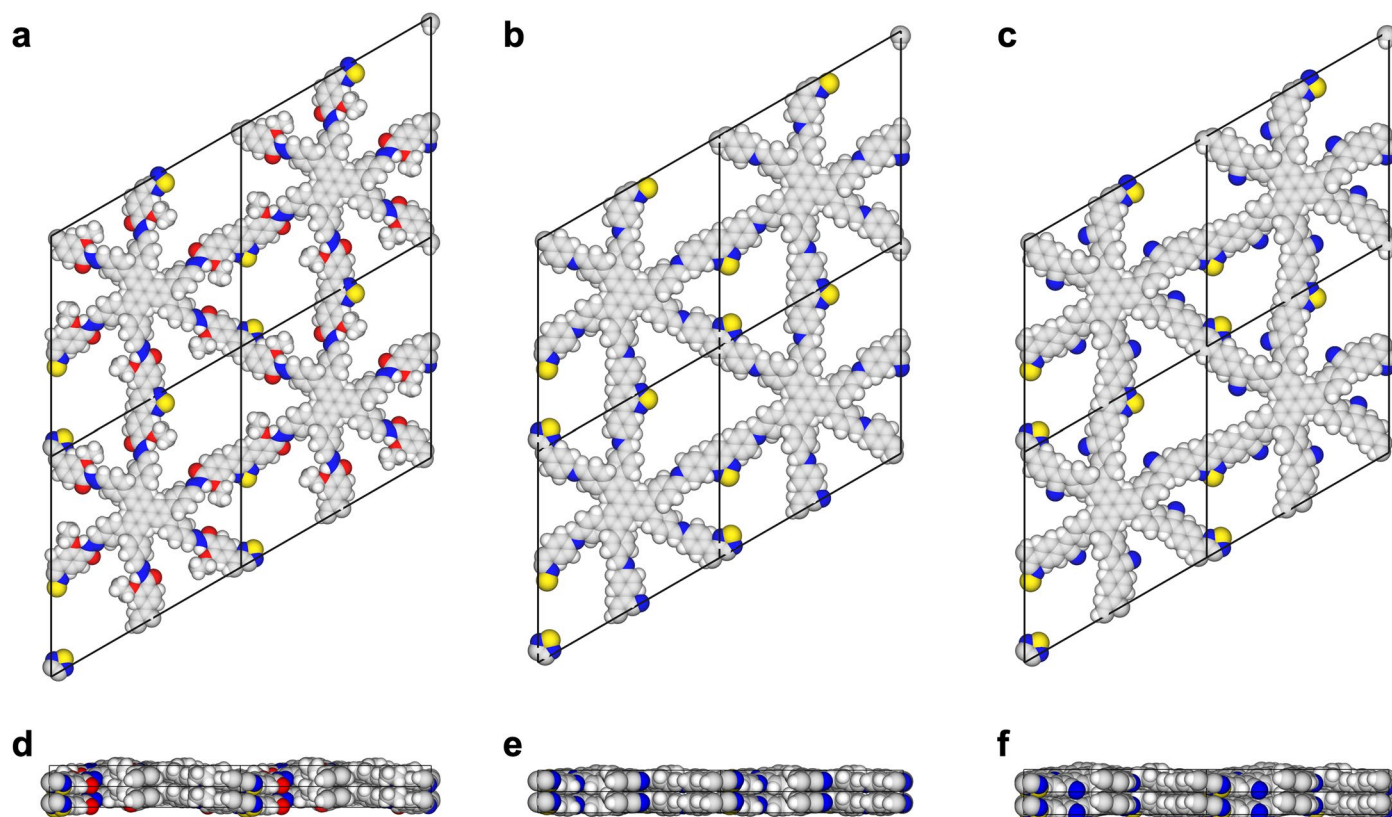
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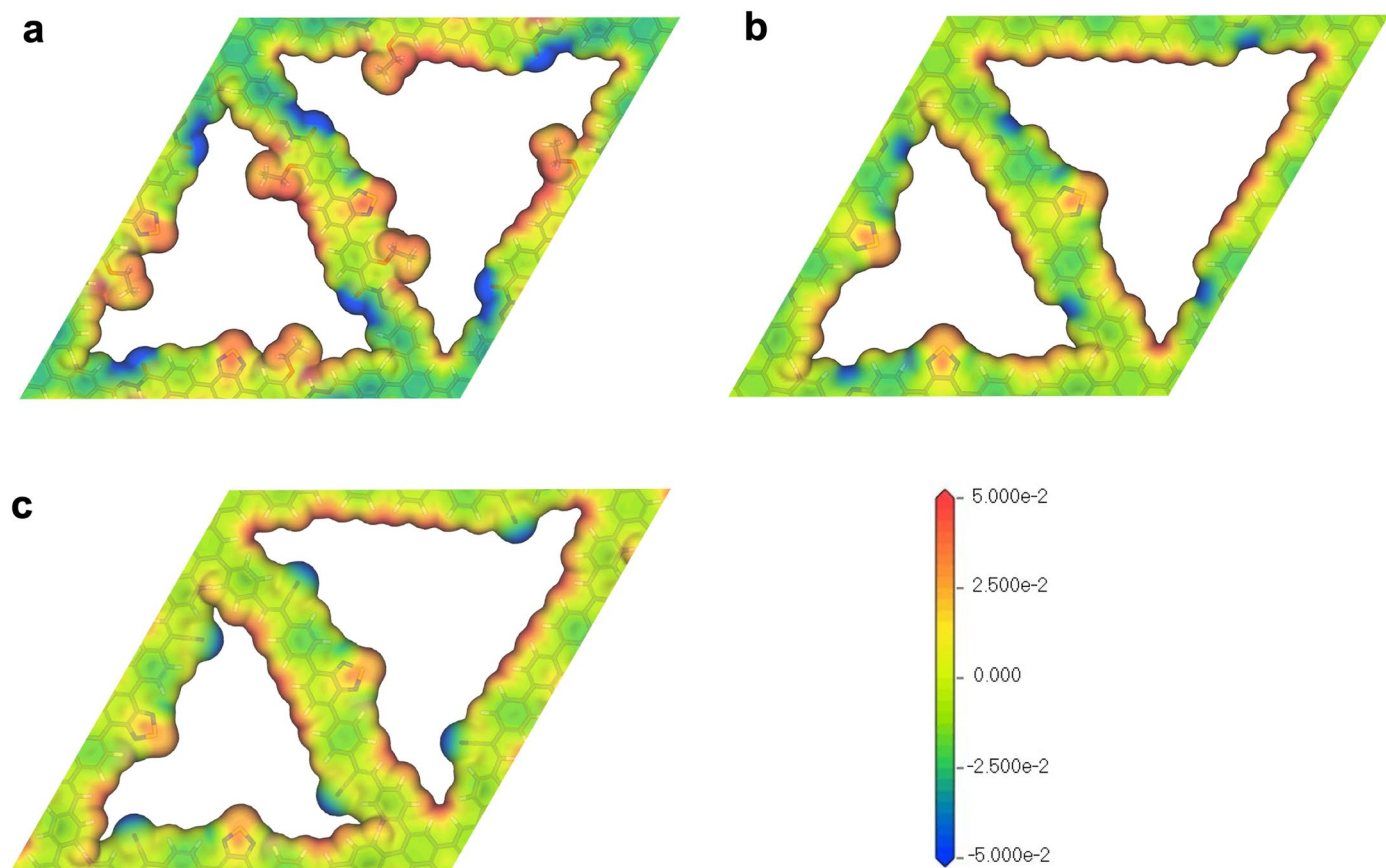
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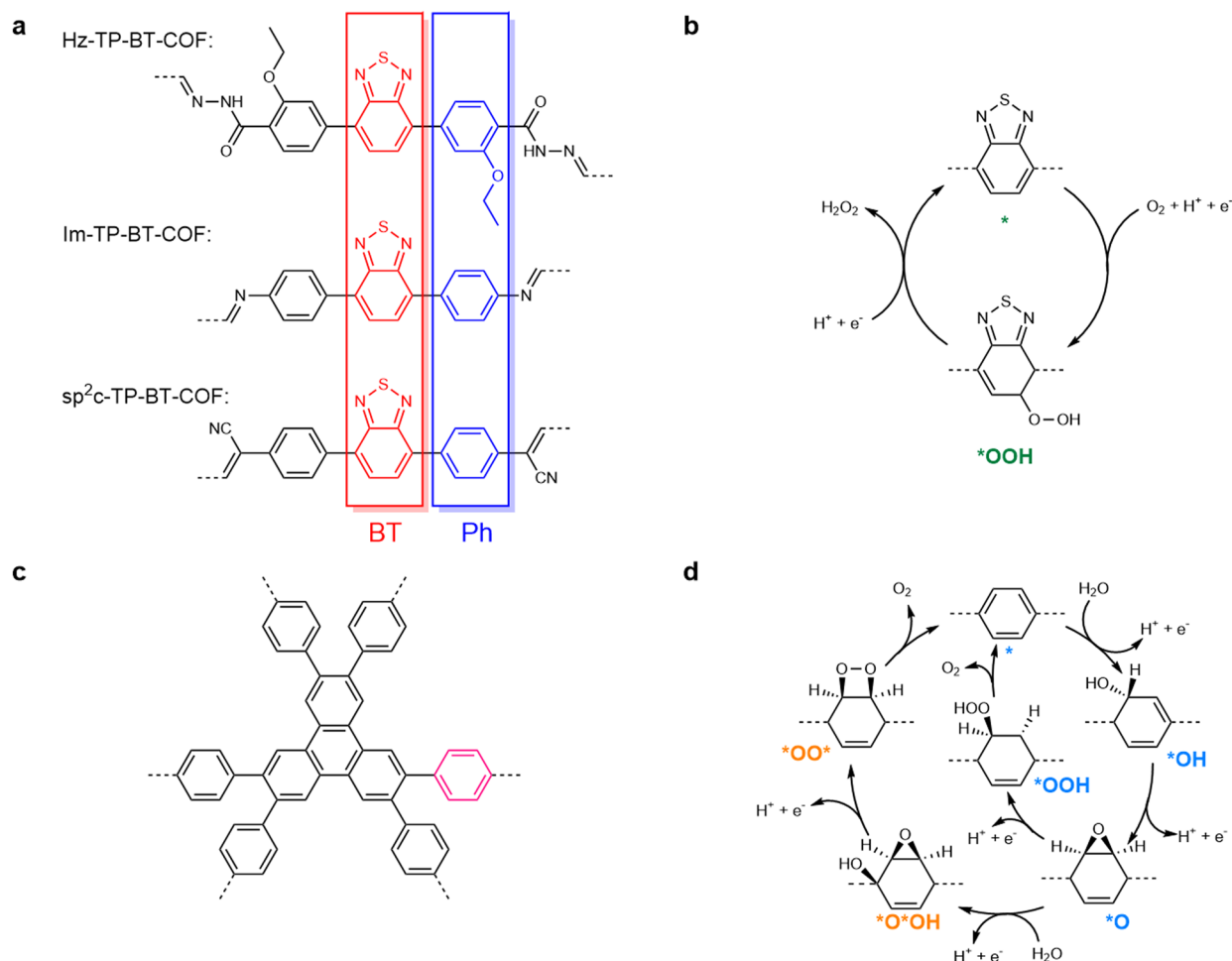
Extended Data Fig. 1 | Powder X-ray diffraction spectra. a, b, Experimentally obtained PXRD patterns (blue cross), Pawley refined patterns (red curves), their differences (grey curves) and simulated sync AA-stacking modes (green curves) of Im-TP-BT-COF (**a**) and sp^2c -TP-BT-COF (**b**).



Extended Data Fig. 2 | Crystal structures. **a–c**, The top view of unit cells of Hz-TP-BT-COF (**a**), lm-TP-BT-COF (**b**) and sp²c-TP-BT-COF (**c**). **d–f**, The side view of unit cells of Hz-TP-BT-COF (**d**), lm-TP-BT-COF (**e**) and sp²c-TP-BT-COF (**f**).



Extended Data Fig. 3 | Electrostatic potentials. a–c, Electrostatic potential map for Hz-TP-BT-COF (**a**) Im-TP-BT-COF (**b**) and sp²c-TP-BT-COF (**c**).



Extended Data Fig. 4 | Active sites and photocatalytic cycles. a, Active sites (BT and Ph) of the three COFs for oxygen reduction reaction. **b,** Photocatalytic cycle for oxygen reduction reaction over the BT unit of benzothiadiazole linker.

c, Active site (Ph unit in pink) for oxygen evolution reaction. **d,** Photocatalytic cycle for oxygen evolution reaction over the Ph unit neighbouring to the triphenylene core.